# Luke Song

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## Education

**University of California, Los Angeles** — Los Angeles, CA
*Bachelor of Science in Computer Science* — *Expected Jun. 2030*

- Courses: Data Structures & Algorithms, Computer Organization, Logic Design of Digital Systems, Calculus of Several Variables.

## Projects

**Emergent Colony** | *C#, Unity 2D* — Jan 2026 - May 2026

- Senior research project. Created simulation of ant colony against evolving predators.
- Implemented pheromone-based distributed decision-making, allowing ants to coordinate using only local environmental information and adapt to static predators.
- Designed defensive and offensive behavioral strategies for ants in response to moving predators.
- Modeled predator evolution under changing ant survival strategies.

**Reinforcement Learning for Connect Four** | *Python, Pytorch, Matplotlib, Kaggle* — Aug 2026

- Developed Reinforcement Learning model for the game of Connect Four.
- Built and optimized custom modular two-part (1) modified monte carlo tree search and (2) ResNet-based dual-head policy and value network.

## Experience

**Research under Prof. H. Blair** — Summer 2024
*University of California, Los Angeles* — *Los Angeles, CA*

- Developed task-driven reinforcement learning model to simulate human running under the effects of ethanol.
- Evaluated and compared three reinforcement learning architectures: DQN, DDPG, and TD3.
- Analyzed and linked simulated behavioral changes to real-world effects of alcohol consumption.

## Awards

**International Logic Olympiad, Grand Champion** — May 2025
*Stanford University* — *Stanford, CA*

- Placed 1st in finals at Stanford University after qualifying among the top 13 teams worldwide.

## Technical Skills

**Languages**: Python, PyTorch, Tensorflow, Numpy, Pandas, JavaScript, HTML/CSS.
**Concepts**: NLP, Transformers, Reinforcement Learning, Deep Learning.
**Developer Tools**: Git, VS Code, Visual Studio, Overleaf, Ollama, Chromadb, Eclipse